

Ihsan Ullah

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TECHNICAL SKILLS

Languages: Python, C++ (basic)

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, CUDA — CNNs, RNNs (LSTM, GRU), Transformers, Reinforcement Learning (PPO), transfer learning, hyperparameter tuning

LLM & NLP: Hugging Face Transformers, LangChain, RAG, Graph RAG, ChromaDB, OpenAI API (GPT-4 Vision), Anthropic API (Claude Vision), vLLM, llama.cpp, BERT, RoBERTa, LoRA/QLoRA fine-tuning, NLTK, SpaCy, Word2Vec, TF-IDF, prompt engineering

Speech & Audio: OpenAI Whisper, librosa (MFCC, chroma, spectral contrast), mel-spectrogram generation, voice fingerprinting, audio segmentation & augmentation

Computer Vision: OpenCV (image processing, contour detection, morphological operations), CNN-based classifiers, YOLO (object detection), OCR (Tesseract / EasyOCR), image augmentation, synthetic image generation

Data Engineering: NumPy, Pandas, Neo4j GraphDB, PostgreSQL, SQL, feature engineering, data preprocessing

Cloud & Deployment: GCP (Compute Engine, Cloud Storage, Secret Manager), Docker, Kubernetes, FastAPI, Pydantic, MLOps pipelines, GPU scheduling, Linux, Git

Visualisation: Matplotlib, Seaborn, Plotly, Streamlit

WORK EXPERIENCE

Associate AI Engineer | Synapsify *May 2026 – Present*

- Working as a Backend AI Engineer on a Unity-integrated AI character creator tool, building both core backend endpoints and Retrieval-Augmented Generation (RAG) endpoints that use prompt engineering and blendshape descriptions to generate characters from a verified dataset of 1,000+ profiles — keeping RAG retrieval logic and general backend services as distinct, purpose-built endpoints.
- Built the RAG pipeline for character data — chunking, dense embedding generation, and vector storage in ChromaDB — to power retrieval-augmented character generation.
- Developed backend endpoints for character save, load, and export operations, backed by a PostgreSQL database and deployed on GCP Compute Engine instances.
- Containerised all backend and RAG services with Docker; as the Associate AI Engineer, used GCP Secret Manager to provision and inject API keys and database credentials into production containers at runtime, keeping secrets out of source code.
- Contributed to a parallel AgriTech initiative: generated synthetic crop imagery to train object detection models for crop identification, with the pipeline planned to extend into disease detection; used GCP Cloud Storage for image and model storage and export.

AI Engineer | ByteBoom *August 2025 – April 2026*

- Architected a Graph RAG cryptocurrency intelligence system built on a Neo4j knowledge graph comprising 32+ typed nodes; developed data-ingestion adapters to retrieve live market data from external APIs and map raw responses into structured graph entities (liquidation zones, whale flows, TA signals, fund flows), enabling context-aware AI reasoning over complex market relationships.
- Designed a Vision-Language Model (VLM) pipeline for automated trading signal confirmation — captured multi-timeframe TradingView chart screenshots and processed them through GPT-4 Vision and Claude Vision APIs for trend detection, support/resistance identification, and candlestick pattern recognition.
- Engineered a Reinforcement Learning (PPO) trading agent integrated with Neo4j GraphDB to extract market-state features, construct state vectors, and optimise action prediction and value estimation through deep RL techniques.
- Built a quantitative trading signals engine incorporating Fibonacci retracements, Elliott Wave theory, Ichimoku Cloud, and Wyckoff methodology for multi-dimensional market analysis and liquidity zone detection.
- Containerised all ML services with Docker, maintaining consistent environments across development and production and enabling one-command deployment.
- Managed collaborative Git workflows with structured branching strategies and merge-review processes.

AI Engineer | University of Peshawar *November 2023 – May 2024*

- Deployed production-grade NLP inference services using FastAPI and Docker on GPU-accelerated Linux servers, supporting real-time model serving at scale.
- Fine-tuned a RoBERTa transformer for fake news classification, achieving a 95% F1-score through systematic architecture search and hyperparameter optimisation.
- Developed a GRU-augmented RoBERTa classifier for e-commerce fake review detection, combining the contextual strength of transformer encoders with recurrent temporal modelling.
- Conducted rigorous model evaluation experiments across multiple architectures, driving incremental accuracy improvements via feature engineering and ablation studies.

PROJECTS

Fake News Detection System — Final Year Project (FYP) *[GitHub]*

- Built a multi-model misinformation classifier comparing a TF-IDF + Logistic Regression baseline against BERT and a RoBERTa + Feedforward Neural Network, reaching a best accuracy of 96%.
- Deployed the classifier as an interactive Streamlit app supporting real-time article submission and side-by-side model comparison.

Quran Reciter Identification & Verse Classification System

- Built an end-to-end audio pipeline: Whisper ASR for transcription, librosa features (MFCC, chroma, spectral contrast, zero-crossing rate) for voice fingerprinting, and a CNN over mel-spectrograms to identify individual reciters.
- Classified verses by thematic content using an LSTM network with Word2Vec embeddings, achieving 91% F1-score.

Graph RAG — Crypto Intelligence System

- Developed ingestion adapters mapping live market API data into a Neo4j knowledge graph (32+ typed nodes covering liquidation zones, whale flows, and fund flows) to support AI-driven market reasoning.
- Combined graph retrieval with LLM reasoning so signals could be explained in natural language rather than raw metrics.

Fake Reviews Detection System

- Built a RoBERTa + GRU classifier for e-commerce review authenticity detection, fusing transformer representations with sequential modelling to reach 93% accuracy.

EDUCATION

BS Computer Science | University of Peshawar *Graduated May 2024*

CERTIFICATIONS

- Machine Learning Specialization — deeplearning.ai / Coursera
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